Lineare Algebra
Serie 24
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Exercise 1

(a)

$$\overline{A+B} = \overline{a_{i,j}} + \overline{b_{i,j}} = \overline{A} + \overline{B}$$
$$\Rightarrow \overline{A+B}^T = \overline{A}^T + \overline{B}^T$$

(b)

$$\lambda \in K$$
$$\overline{\lambda A}`T = \overline{\lambda a_{i,j}}^T = \overline{\lambda}\overline{a_{j,i}} = \overline{\lambda} \cdot \overline{A}^T$$

(c)

$$\overline{\overline{A}^T}^T = \overline{\overline{a_{j,i}}}^T = a_{i,j} = A$$

(d)

$$\overline{I_n}^T = \overline{I_n} = I_n$$

(e)

$$\overline{AC}^T = \overline{\sum_{k=1}^{m} a_{i,k} c_{k,j}}^T = \sum_{k=1}^{m} \overline{a_{i,k} c_{k,j}}^T = \sum_{k=1}^{m} \overline{c_{j,k}} \cdot \overline{a_{k,i}} = \overline{C}^T \cdot \overline{A}^T$$

Exercise 2

$$\langle p, q \rangle = \int_0^1 p(t)q(t)dt$$

(a)

Find an orthonormal basis of the vector space of span$\{1, x, x^2\}$

$$\begin{aligned}
p_1(x) &= 1 \\
p_2(x) &= x - \langle 1, x \rangle 1 = x - \int_0^1 1x dx = x - \frac{1}{2} \\
p_3(x) &= x^2 - \langle 1, x^2 \rangle 1 - \langle x - \frac{1}{2}, x^2 \rangle \left(x - \frac{1}{2}\right) \\
&= x^2 - \int_0^1 1x^2 dx - \left(\int_0^1 x^3 dx - \int_0^1 \frac{1}{2}x^2\right)\left(x - \frac{1}{2}\right) \\
&= x^2 - \frac{1}{3} - \left(x - \frac{1}{2}\right)\left(\frac{1}{4} - \frac{1}{6}\right) \\
&= x^2 - \frac{1}{12}x - \frac{7}{24}
\end{aligned}$$

$$\|p_1\| = \sqrt{\langle 1\ ,\ 1\rangle} = 1$$

$$\|p_2\| = \sqrt{\langle x - \frac{1}{2}\ , x - \frac{1}{2}\rangle} = \sqrt{\int_0^1 \left(x - \frac{1}{2}\right)^2 dx} = \frac{1}{2\sqrt{3}}$$

$$\|p_3\| = \sqrt{\langle x^2 - \frac{1}{12}x - \frac{7}{24}\ ,\ x^2 - \frac{1}{12}x - \frac{7}{24}\rangle} = \sqrt{\int_0^1 \left(x^2 - \frac{1}{12}x - \frac{7}{24}\right)^2 dx} = \frac{\sqrt{9795}}{360}$$

$$e_1 = \frac{p_1}{\|p_1\|} = 1$$

$$e_2 = \frac{p_2}{\|p_2\|} = 2\sqrt{3}\left(x - \frac{1}{2}\right) = 2\sqrt{3}x - \sqrt{3}$$

$$e_3 = \frac{p_3}{\|p_3\|} = \frac{360}{\sqrt{9795}}\left(x^2 - \frac{1}{12}x - \frac{7}{24}\right)$$

(b)

Take the basis from (a) and apply the differentiation operator D to it.

$$D(e_1) = 0$$

$$D(e_2) = 2\sqrt{3} = 2\sqrt{3}e_1$$

$$D(e_3) = \frac{360}{\sqrt{9795}}\left(2x - \frac{1}{12}\right) = \frac{720}{\sqrt{9795}}x - \frac{30}{\sqrt{9795}} = \frac{360}{\sqrt{9795}}e_2 + \frac{330}{\sqrt{9795}}e_1$$

$$\Rightarrow D = \begin{pmatrix} 0 & 2\sqrt{3} & \frac{330}{\sqrt{9795}} \\ 0 & 0 & \frac{360}{\sqrt{9795}} \\ 0 & 0 & 0 \end{pmatrix}$$

Minimizing the distance to a subset.

let $p = P_U(v) \in U$

using pythagorean theorem

$$\|v - u\|^2 = \|v - p\|^2 + \|p - u\|^2 \geq \|v - p\|^2$$

$$\Rightarrow \|v - p\| = \|v - P_U(v)\| \leq \|v - u\|$$

$u = P_U(v)$

$$\Rightarrow \|v - P_U(v)\| = \|v - u\| \quad \forall u \in U$$

Exercise 4

Let $K = \mathbb{R}$. On $K[x]_5$ with the inner product

$$\langle p, q\rangle = \int_{-\pi}^{\pi} p(t)q(t)dt$$

let $p(x) = ax^5 + bx^4 + cx^3 + dx^2 + ex + f \in K[x]_5$

We want to minimize the integral

$$\|\sin - p\|^2 = \langle \sin - p, \sin - p\rangle = \int_{-\pi}^{\pi} |\sin(t) - p(t))|^2\ dt$$

For this we need to compute the projection of sin onto $K[x]_5$
We can take the standard basis of $K[x]_5$

$$B = (1, x, x^2, x^3, x^4, x^5)$$

$$\Rightarrow \langle \sin - p, 1 \rangle = \int_{-\pi}^{\pi} (\sin(t) - p(t))dt = 0$$

$$\Rightarrow \langle \sin - p, x \rangle = \int_{-\pi}^{\pi} (\sin(t) - p(t))tdt = 0$$

$$\Rightarrow \langle \sin - p, x^2 \rangle = \int_{-\pi}^{\pi} (\sin(t) - p(t))t^2dt = 0$$

$$\Rightarrow \langle \sin - p, x^3 \rangle = \int_{-\pi}^{\pi} (\sin(t) - p(t))t^3dt = 0$$

$$\Rightarrow \langle \sin - p, x^4 \rangle = \int_{-\pi}^{\pi} (\sin(t) - p(t))t^4dt = 0$$

$$\Rightarrow \langle \sin - p, x^5 \rangle = \int_{-\pi}^{\pi} (\sin(t) - p(t))t^5dt = 0$$

We can ignore the odd powers of x, since the product of an odd and an even function is odd and the integral of an odd function over a symmetric interval is zero.
We get the following system of equations:

$$\int_{-\pi}^{\pi} (\sin(t) - ax^5 - cx^3 - ex)dt = 0$$

$$\int_{-\pi}^{\pi} (\sin(t) - ax^5 - cx^3 - ex)x^2dt = 0$$

$$\int_{-\pi}^{\pi} (\sin(t) - ax^5 - cx^3 - ex)x^4dt = 0$$

we can compute the integrals:

$$\int_{-\pi}^{\pi} t\sin(t)dt = 2\pi$$

$$\int_{-\pi}^{\pi} t^3\sin(t)dt = 2\pi^3 - 12\pi$$

$$\int_{-\pi}^{\pi} t^5\sin(t)dt = 2\pi^5 - 40\pi^3 + 240\pi$$

$$\Rightarrow$$

$$\frac{1}{6}a\pi^6 + \frac{1}{4}c\pi^4 + \frac{1}{2}e\pi^2 = 2\pi$$

$$\frac{1}{8}a\pi^8 + \frac{1}{6}c\pi^6 + \frac{1}{4}e\pi^4 = 2\pi^3 - 12\pi$$

$$\frac{1}{10}a\pi^{10} + \frac{1}{8}c\pi^8 + \frac{1}{6}e\pi^6 = 2\pi^5 - 40\pi^3 + 240\pi$$

using Calculator we get:

$$\dots$$

Exercise 5

Proof by counterexample:
define the function f_ε

$$f_\varepsilon(x) = \begin{cases} 0 & x \in [-1{,}1] \setminus [-\varepsilon{,}\varepsilon] \\ \frac{\varepsilon - |x|}{\varepsilon} & x \in [-\varepsilon \ , \ \varepsilon] \end{cases}$$

$$\begin{aligned} &\Rightarrow \varphi(f_\varepsilon) = 1 \quad \forall \varepsilon \\ &\int_{-1}^{1} f(x)g(x)dx = \int_{-\varepsilon}^{\varepsilon} f(x)g(x)dx \\ &\qquad\qquad\qquad\quad \leq M2\varepsilon \end{aligned}$$

The function g is continuous therefore $\exists M = \max(g(x)), x \in [-1, 1]$
Set ε such that $2M\varepsilon < 1$. This means for every function g we can find a function f_ε contradicting the statement.

Exercise 6

(a)

$T^\star$ is the adjoint operator of T if

$$\forall v \in V, w \in W : \langle T(v), w\rangle_W = \langle v, T^\star(w)\rangle_V$$

Let $\varphi, T(v) \in \mathbb{R}^E$

$$\begin{aligned} \langle T(f), \varphi\rangle_E &= \sum_{e=(u,v)\in E} (f(v) - f(u))\varphi(e) \\ &= \sum_{e=(u,v)\in E} \varphi(e)f(v) - \sum_{e=(u,v)\in E} \varphi(e)f(u) \\ &= \sum_{v\in V} f(v)\left(\sum_{e=(u,v)} \varphi(u,v) - \sum_{e=(v,u)} \varphi(v,u)\right) \\ &= \sum_{v\in V} f(v)S(\varphi)(v) \\ &= \langle f, S(\varphi)\rangle_V = \langle f, T^\star(\varphi)\rangle_V \end{aligned}$$

□

Let $e \in E$

$$T^\star T(f)(e) = \sum_{(u,v)\in E} (f(v) - f(u)) - \sum_{(v,u)\in E} (f(u) - f(v))$$

(b)

Define the basis as all vertices of the graph.
Then we can compute the matrix of $T^\star T$ with respect to this basis.
Let v_i be the i-th vertex of the graph.
using the result from part (a) we have:

$$
\begin{aligned}
T^{\star}T(v_i) &= \sum_{(u,v_i)\in E} (v_i - u) - \sum_{(v_i,u)\in E} (u - v_i) \\
&= \sum_{(u,v_i)\in E} v_i - \sum_{(u,v_i)\in E} u - \sum_{(v_i,u)\in E} u + \sum_{(v_i,u)\in E} v_i \\
&= \deg(v_i)v_i - \sum_{(u,v_i)\in E} u - \sum_{(v_i,u)\in E} u + \deg(v_i)v_i \\
&= 2\deg(v_i)v_i - \sum_{(u,v_i)\in E} u - \sum_{(v_i,u)\in E} u
\end{aligned}
$$

since the graph is undirected and $\deg(v_i) = d$ we have

$$
\begin{aligned}
&\Rightarrow T^{\star}T(v_i) = 2dv_i - 2\sum_{u\sim v_i} u \\
&\Rightarrow T^{\star}T = 2dI - 2A \\
&\Rightarrow (T^{\star}T)_{i,j} = \begin{cases} 2d & i = j \\ -2 & i \sim j \\ 0 & \text{otherwise} \end{cases}
\end{aligned}
$$

Since the the Graph is undirected $\Rightarrow A = A^T$. The Graph is d- regular, this means all vertices has degree d. Therefore the sum of ech row and columm of A is d.

$$
\begin{aligned}
&\Rightarrow \sum_{j=1}^{n} Id - A_{i,j} = d - d = 0 \quad \forall i \in \{1, ..., n\} \\
&\Rightarrow (T^{\star}T)\begin{pmatrix} 1 \\ 1 \\ \vdots \\ 1 \end{pmatrix} = 2(Id - A)\begin{pmatrix} 1 \\ 1 \\ \vdots \\ 1 \end{pmatrix} = 0
\end{aligned}
$$

$T^{\star}T$ admits 0 as an eigenvalue.

Single Choice

1. b
2. c

Multiple Choice Fragen

a and b are correct.